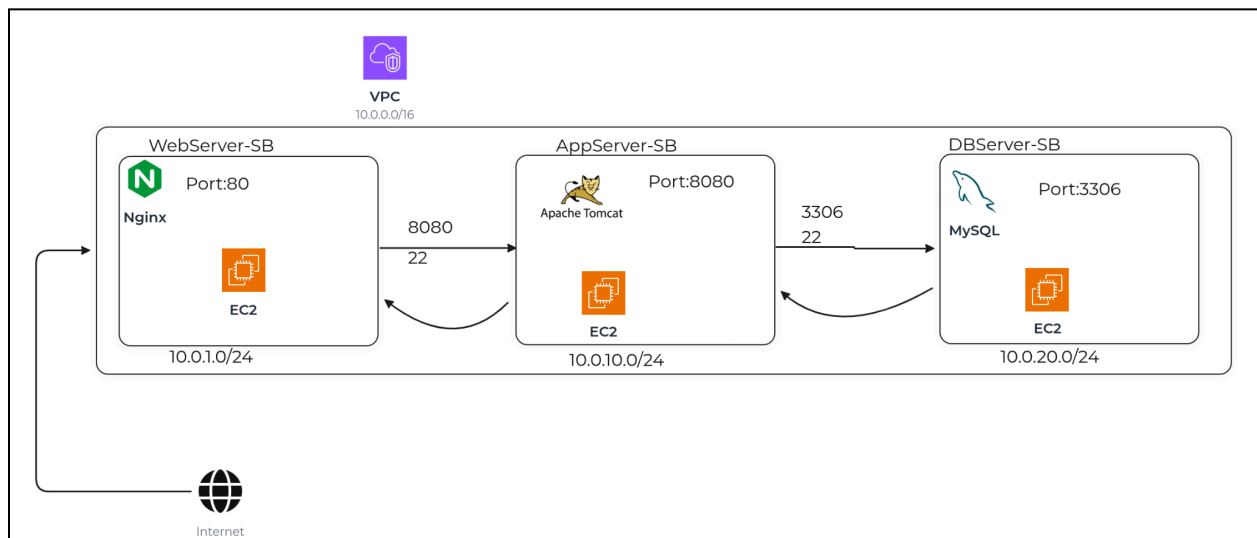


Mohammed Abdul Sami
Batch-18

3-Tier Architecture

Architecture



CONFIGURATION SCREENSHOT

EC2 > Instances

Instances (3) Info Last updated about 1 hour ago Connect Instance state Actions Launch inst...

Saved filter sets
Choose filter set Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...
☐	DB-server	i-0fef014ffa8477585	Running	t3.micro	3/3 checks passed	ap-south-2b	-	-
☐	Webserver	i-093b5fb717b96bb99	Running	t3.micro	3/3 checks passed	ap-south-2a	-	18.60.42.193
☐	AppServer	i-033b5976ef701ad83	Running	t3.micro	3/3 checks passed	ap-south-2a	-	-

0597494146bb9040c

rtb-0597494146bb9040c / Default-PUB-RT

Actions

Details Info

Route table ID
rtb-0597494146bb9040c

VPC
vpc-0a4876798664ea3c2 | HYD1-VPC

Main
Yes

Owner ID
969079689686

Explicit subnet associations
3 subnets

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	lgw-02fc779e5d9fd0ab7	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

rtb-0ada5e1432937e2af / Appserver-Pvt-SG

Actions

Details Info

Route table ID
rtb-0ada5e1432937e2af

VPC
vpc-0a4876798664ea3c2 | HYD1-VPC

Main
No

Owner ID
969079689686

Explicit subnet associations
-

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1181ce51687f4c744	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

rtb-0655a01a6c4b7b5cf / db-server-RT

Actions

Details Info

Route table ID
rtb-0655a01a6c4b7b5cf

VPC
vpc-0a4876798664ea3c2 | HYD1-VPC

Main
No

Owner ID
969079689686

Explicit subnet associations
3 subnets

Edge associations
-

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Both Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1181ce51687f4c744	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

16.113.10.28

Actions Associate Elastic IP address

Summary

Allocated IPv4 address

16.113.10.28

Association ID

elpassoc-0a21b5b0cab7a23d5

Network interface ID

-

Address pool

Amazon

Type

Public IP

Scope

VPC

Network interface owner account ID

-

Service managed

-

Allocation ID

eipalloc-078cfa5a3bcb71dcd

Associated instance ID

-

Public DNS

-

Reverse DNS record

-

Private IP address

16.113.10.28

NAT Gateway ID

nat-1181ce51687f4c744 (NAT-Gw)

Tags(0)

Manage tags

sg-093ec075f1680faaa - default

Action

Details

Security group name

default

Security group ID

sg-093ec075f1680faaa

Description

default VPC security group

VPC ID

vpc-0a4876798664ea3c2

Owner

969079689686

Inbound rules count

3 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (3)

Manage tags Edit inbound rule

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-06b7210796b6a90c9	-	All traffic	All	All	sg-093ec075f1680faaa ...
sgr-04eb96ce650814ba7	IPv4	HTTP	TCP	80	0.0.0.0/0
sgr-04f54bef378c9d664	IPv4	SSH	TCP	22	0.0.0.0/0

sg-046edc8229ddd2f81 - Pvt-SG

Actions

Details

Security group name

Pvt-SG

Security group ID

sg-046edc8229ddd2f81

Description

Pvt-SG

VPC ID

vpc-0a4876798664ea3c2

Owner

969079689686

Inbound rules count

2 Permission entries

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing

VPC associations

Related resources - new

Tags

Inbound rules (2)

Manage tags Edit inbound rules

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-0818a684ffa0c45e0	IPv4	SSH	TCP	22	0.0.0.0/0
sgr-0bb7c92b0616cd764	IPv4	All traffic	All	All	10.0.1.116/32

sg-046edc8229ddd2f81 - Pvt-SG

Details

Security group name Pvt-SG	Security group ID sg-046edc8229ddd2f81	Description Pvt-SG	VPC ID vpc-0a4876798664ea3c2
Owner 969079689686	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | **Outbound rules** | **Sharing** | **VPC associations** | **Related resources - new** | **Tags**

Outbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0f2d3d6d264e7da11	IPv4	All traffic	All	All

sg-0654c00272a6acd97 - dbserver-SG

Details

Security group name dbserver-SG	Security group ID sg-0654c00272a6acd97	Description dbserver-SG	VPC ID vpc-0a4876798664ea3c2
Owner 969079689686	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | **Outbound rules** | **Sharing** | **VPC associations** | **Related resources - new** | **Tags**

Inbound rules (2)

Search

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-099de84cd6f1513d9	IPv4	SSH	TCP	22	10.0.10.188/32
sgr-08c7b78197101a35c	IPv4	MYSQL/Aurora	TCP	3306	10.0.10.188/32

sg-0654c00272a6acd97 - dbserver-SG

Details

Security group name dbserver-SG	Security group ID sg-0654c00272a6acd97	Description dbserver-SG	VPC ID vpc-0a4876798664ea3c2
Owner 969079689686	Inbound rules count 2 Permission entries	Outbound rules count 1 Permission entry	

Inbound rules | **Outbound rules** | **Sharing** | **VPC associations** | **Related resources - new** | **Tags**

Outbound rules (1)

Search

Name	Security group rule ID	IP version	Type	Protocol	Port range
-	sgr-0b19b0bffd4c45edb3	IPv4	All traffic	All	All

Setting Up the Webserver (Nginx)

1. Created EC2 instance and installed nginx & telnet package utilizing the Public Subnet attached to the Internet Gateway

VPC > Route tables > rtb-0597494146bb9040c > Edit routes

Edit routes: rtb-0597494146bb9040c / Default-PUB-RT

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/16	local	Active	No	CreateRouteTable
0.0.0.0/0	Internet Gateway	Active	No	CreateRoute

Add route

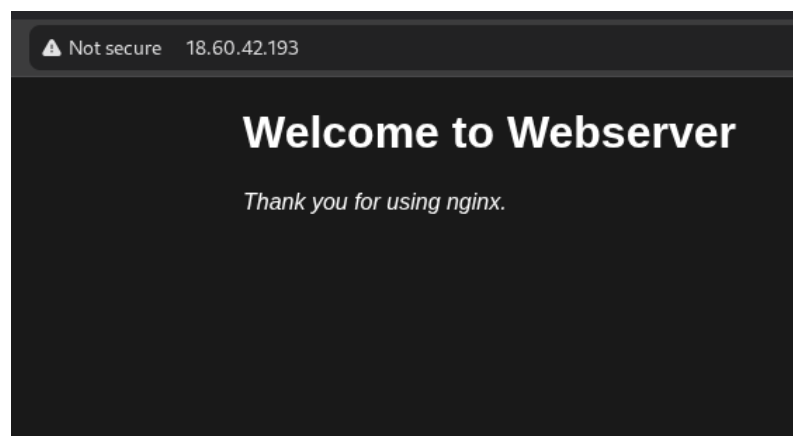
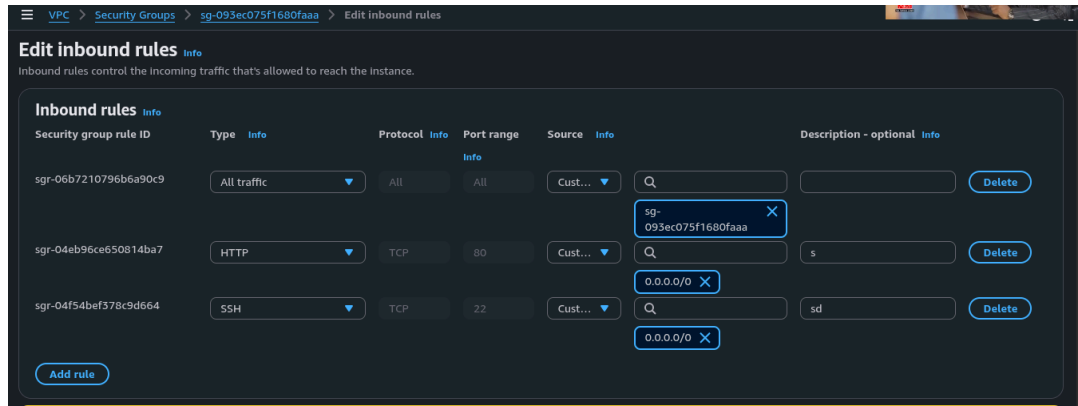
Sudo apt update && sudo apt install nginx telnet -y

2. Edit the nginx default index.html file in the /var/www/html/
3. Started the nginx server `sudo systemctl enable nginx`, `sudo systemctl start nginx`
4. Checked the status using `sudo systemctl status nginx`

```
ubuntu@webserver:~$ sudo systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-30 05:44:03 UTC; 8min ago
     Invocation: 396612d02f244659953ddc95788bd457
       Docs: man:nginx(8)
    Main PID: 1744 (nginx)
      Tasks: 3 (limit: 627)
     Memory: 3M (peak: 7.3M)
        CPU: 43ms
    CGroup: /system.slice/nginx.service
            └─1744 "nginx: master process /usr/sbin/nginx -g daemon on; master_proces
              └─1747 "nginx: worker process"
                └─1748 "nginx: worker process"

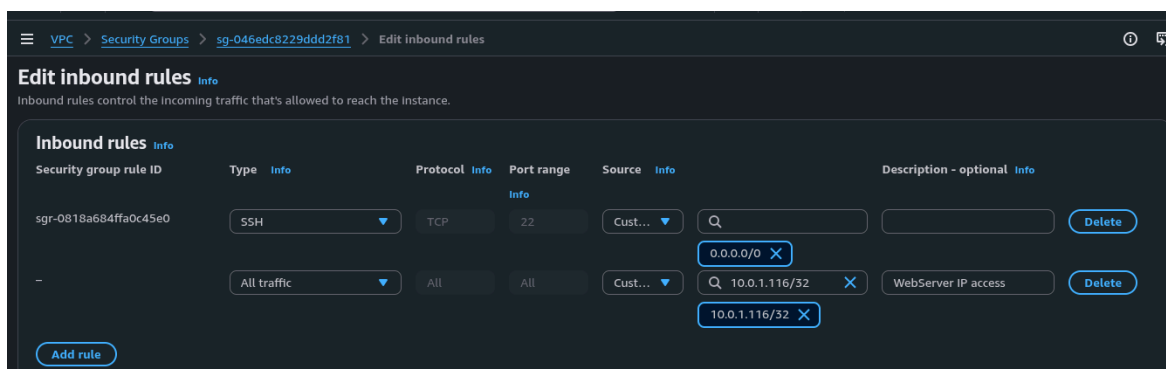
Aug 30 05:44:03 webserver systemd[1]: Starting nginx.service - A high performance web
Aug 30 05:44:03 webserver systemd[1]: Started nginx.service - A high performance web s
ubuntu@webserver:~$
```

5. I have added an HTTP allow rule to any IPv4 address Inbound rule to the Security Group

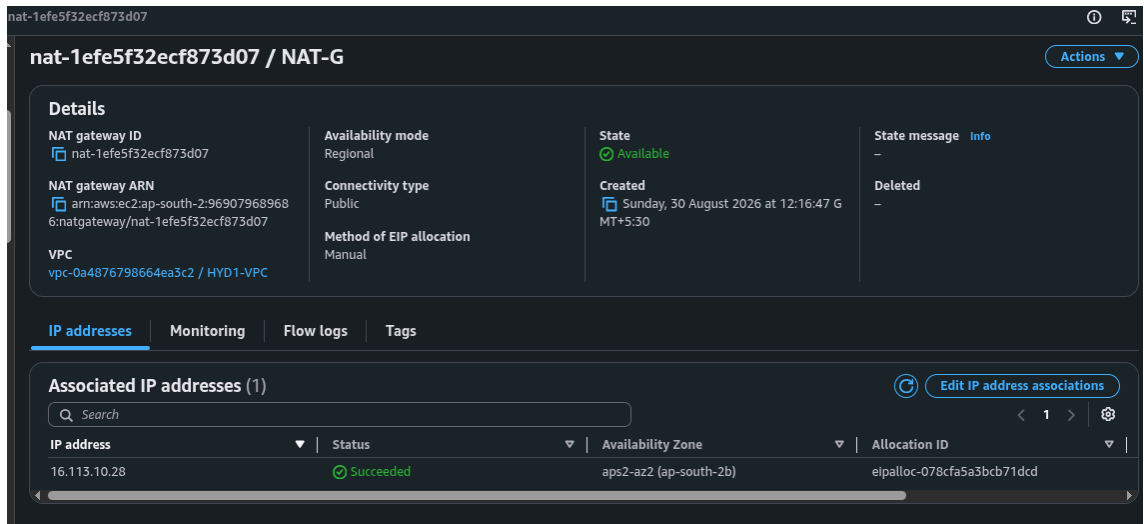


Setting Up the Application Server (Tomcat)

1. Created Ec2 instance in the private subnet with **Auto-assign public IP** to Disable
2. Added a inbound rule to allow only WebServer's IP for the communication.



3. For the installation of the Tomcat, I will require creating the Elastic IP address & Nat Gateway will be assigned it for the internet access via the NAT Gateway.



4. Adding the route the private subnets route tables to link it with the Nat Gateway.



5. To SSH into the Private server (AppServer) via the Public Webserver server. We need to login to Public Webserver and then do the SSH but here I will be using a different method (Jump Proxy).

```
~ /Downloads 12:36:18 PM
ssh -i KaliHyd.pem -J ubuntu@18.60.42.193 ubuntu@10.0.10.188
The authenticity of host '10.0.10.188 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is: SHA256:H9F9LvRCGn+yJ2bpfmSVYpUamwNzYJgopL5W70LGQmg
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes_
```

```
~ /Downloads 0113:01 [32/34]
ssh -i KaliHyd.pem -J ubuntu@18.60.42.193 ubuntu@10.0.10.188
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

* Documentation:  https://docs.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Sun Aug 30 07:30:27 UTC 2026

System load:  0.0           Temperature:   -273.1 C
Usage of /:   30.7% of 6.61GB Processes:    118
Memory usage: 28%          Users logged in: 0
Swap usage:   0%           IPv4 address for ens5: 10.0.10.188

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
Failed to connect to https://changelogs.ubuntu.com/meta-release. Check your
Internet connection or proxy settings

Last login: Sun Aug 30 07:07:38 2026 from 10.0.1.116
ubuntu@appserver:~$
```

6. As I have connected the NAT gateway, let me verify if the internet is reaching or not. `ping [google.com](https://www.google.com)`

```
ubuntu@appserver:~$ ping google.com -c 3
PING google.com (142.250.183.174) 56(84) bytes of data.
64 bytes from bom07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=1 ttl=115 time=26.1
64 bytes from bom07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=2 ttl=115 time=25.9
64 bytes from bom07s32-in-f14.1e100.net (142.250.183.174): icmp_seq=3 ttl=115 time=25.9

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2004ms
rtt min/avg/max/mdev = 25.874/25.949/26.092/0.100 ms
ubuntu@appserver:~$ sudo apt update
Hit:1 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
119 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@appserver:~$
```

7. Since, we got the internet access via the NAT network, now let's install tomcat

```
ubuntu@appserver:~$ wget https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.59/bin/apache-tomcat-10.1.59.zip
--2026-08-30 08:15:51-- https://dlcdn.apache.org/tomcat/tomcat-10/v10.1.59/bin/apache-tomcat-10.1.59.zip
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)|151.101.2.132|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14977703 (14M) [application/zip]
Saving to: 'apache-tomcat-10.1.59.zip'

apache-tomcat-10.1.59.zip      100%[=====] 14.28M
2026-08-30 08:15:51 (371 MB/s) - 'apache-tomcat-10.1.59.zip' saved [14977703/14977703]

ubuntu@appserver:~$ ls
apache-tomcat-10.1.59.zip
ubuntu@appserver:~$ unzip apache-tomcat-10.1.59.zip
Archive:  apache-tomcat-10.1.59.zip
  creating: apache-tomcat-10.1.59/
  creating: apache-tomcat-10.1.59/bin/
  creating: apache-tomcat-10.1.59/conf/
```

8. We need Java for running tomcat, so, we need to install openjdk-25

```
ubuntu@appserver:~$ sudo apt install openjdk-25-jdk -y
Installing:
  openjdk-25-jdk

Installing dependencies:
  adwaita-icon-theme  libaom3  libgbm1
  alsa-topology-conf  libasound2-data  libgdk-pixbuf-2.0-
  alsa-ucm-conf       libasound2t64   libgdk-pixbuf2.0-c
  at-spi2-common      libatk-bridge2.0-0t64  libgif7
  at-spi2-core        libatk-wrapper-java  libgl1
  bubblewrap          libatk-wrapper-java-jni  libgl1-mesa-dri
```

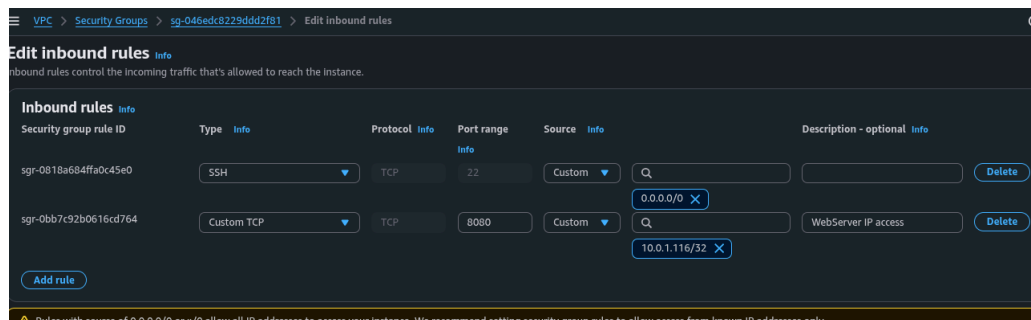
9. I have cd to the directory to start the tomcat server. Tomcat server started

```
ubuntu@appserver:~$ ls
apache-tomcat-10.1.59  apache-tomcat-10.1.59.zip
ubuntu@appserver:~$ cd apache-tomcat-10.1.59/
ubuntu@appserver:~/apache-tomcat-10.1.59$ ls
BUILDING.txt  CONTRIBUTING.md  LICENSE  NOTICE  README.md  RELEASE-NOTES  RUNNING.txt  bin  conf  lib  logs  t
ubuntu@appserver:~/apache-tomcat-10.1.59$ cd bin/
ubuntu@appserver:~/apache-tomcat-10.1.59/bin$ ls
bootstrap.jar  ciphers.bat  configtest.bat  digest.sh  migrate.sh  shutdown.sh
catalina-tasks.xml  ciphers.sh  configtest.sh  makebase.bat  setclasspath.bat  startup.bat
catalina.bat  commons-daemon-native.tar.gz  daemon.sh  makebase.sh  setclasspath.sh  startup.sh
catalina.sh  commons-daemon.jar  digest.bat  migrate.bat  shutdown.bat  tomcat-juli.
ubuntu@appserver:~/apache-tomcat-10.1.59/bin$ chmod +x startup.sh
ubuntu@appserver:~/apache-tomcat-10.1.59/bin$ ./startup.sh
Cannot find ./catalina.sh
The file is absent or does not have execute permission
This file is needed to run this program
ubuntu@appserver:~/apache-tomcat-10.1.59/bin$ chmod +x catalina.sh
ubuntu@appserver:~/apache-tomcat-10.1.59/bin$ ./startup.sh
Using CATALINA_BASE:   /home/ubuntu/apache-tomcat-10.1.59
Using CATALINA_HOME:   /home/ubuntu/apache-tomcat-10.1.59
Using CATALINA_TMPDIR: /home/ubuntu/apache-tomcat-10.1.59/temp
Using JRE_HOME:        /usr
Using CLASSPATH:        /home/ubuntu/apache-tomcat-10.1.59/bin/bootstrap.jar:/home/ubuntu/apache-tomcat-10.1.59
Using CATALINA_OPTS:
Tomcat started.
ubuntu@appserver:~/apache-tomcat-10.1.59/bin$
```

10. Now let's comment out the <valve> tags in the context.xml in host-manager & manager directories. So, we don't have any IP restrictions.

```
ubuntu@appserver:~/apache-tomcat-10.1.59$ sudo vi webapps/
ROOT/      docs/      examples/  host-manager/  manager/
ubuntu@appserver:~/apache-tomcat-10.1.59$ sudo vi webapps/manager/
META-INF/  WEB-INF/  css/      images/  index.jsp  status.xsd  xform.xml
ubuntu@appserver:~/apache-tomcat-10.1.59$ sudo vi webapps/manager/
META-INF/  WEB-INF/  css/      images/  index.jsp  status.xsd  xform.xml
ubuntu@appserver:~/apache-tomcat-10.1.59$ sudo vi webapps/manager/META-INF/context.xml
ubuntu@appserver:~/apache-tomcat-10.1.59$ sudo vi webapps/host-manager/META-INF/context.xml
ubuntu@appserver:~/apache-tomcat-10.1.59$
```

11. I had to edit the old rule in SG group of the private subnet attached to Application Server EC2 instance and only allow the TCP port 8080 to webserver IP.



```
ubuntu@appserver:~/apache-tomcat-10.1.59$ ip a | grep inet
inet 127.0.0.1/8 scope host lo
inet6 ::1/128 scope host noprefixroute
inet 10.0.10.188/24 metric 100 brd 10.0.10.255 scope global dynamic ens5
inet6 fe80::483:ecff:fed3:1f17/64 scope link proto kernel_ll
ubuntu@appserver:~/apache-tomcat-10.1.59$

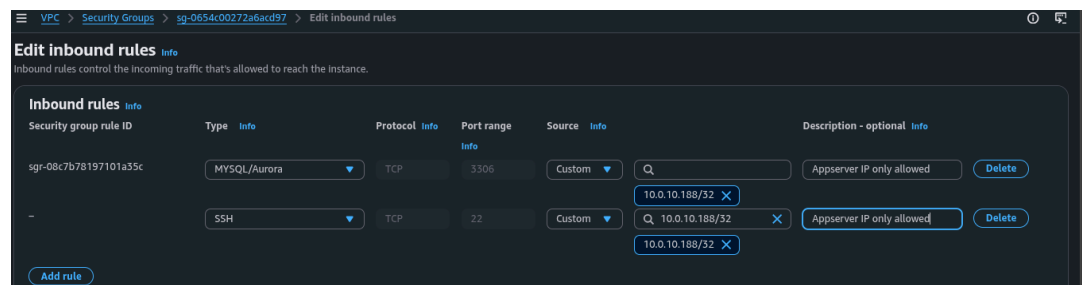
ubuntu@webserver:~$ telnet 10.0.10.188 8080
Trying 10.0.10.188...
Connected to 10.0.10.188.
Escape character is '^]'.
Connection closed by foreign host.
ubuntu@webserver:~$
```

12. We have successfully connected via telnet <appserver-ip> 8080. Hence connection establish connection between WebServer (Nginx) to Application

```
ubuntu@webserver:~$ telnet 10.0.10.188 8080
Trying 10.0.10.188...
Connected to 10.0.10.188.
Escape character is '^]'.
Connection closed by foreign host.
ubuntu@webserver:~$ curl -I http://10.0.10.188:8080
HTTP/1.1 200
Content-Type: text/html; charset=UTF-8
Date: Sun, 30 Aug 2026 09:08:20 GMT
ubuntu@webserver:~$
```

Setting Up the DB Server (MySQL)

1. Created Ec2 instance in the private subnet with **Auto-assign public IP** to Disable
2. Create a new SG group added an inbound rule to allow only AppServer's IP for the communication. And only port 3306 is allowed. Also, ssh from AppServer's IP.



3. I'm trying to connect SSH via the Jump proxy method to the Internal Private IP.

```
ssh -i KaliHyd.pem -J ubuntu@18.60.42.193,ubuntu@10.0.10.188 ubuntu@10.0.20.176
The authenticity of host '10.0.20.176 (<no hostip for proxy command>)' can't be established.
ED25519 key fingerprint is: SHA256:NrbSYtcwBR83rOfToeIsWg0HVhcSpU6L57vls7pQYuc
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
```

4. Installing the MYSQL server using apt manager.

```
ubuntu@dbserver:~$ sudo apt install mysql -y
Error: Unable to locate package mysql
ubuntu@dbserver:~$ sudo apt install mysql-server -y
Installing:
  mysql-server

Installing dependencies:
  libgoogle-perftools4t64  libprotobuf-lite32t64  mysql
  libmecab2                libtcmalloc-minimal4t64  mysql
```

5. Secure installation using the command `sudo mysql\_secure\_installation`. If it prompts message read the prompt and type y/n as per your requirement.

```
ubuntu@dbserver:~$ sudo mysql_secure_installation

Securing the MySQL server deployment.

Connecting to MySQL using a blank password.
The 'validate_password' component is installed on the server.
The subsequent steps will run with the existing configuration
```

6. Edit configuration file at /etc/mysql/mysql.conf.d/mysqld.cnf to 0.0.0.0 or AppServer's IP address.

```
# Instead of skip-networking the default is now to listen only on
# localhost which is more compatible and is not less secure.
bind-address            = 0.0.0.0
#bind-addrees to 0.0.0.0 or <Appserve IPv4>
mysqlx-bind-address     = 127.0.0.1
```

7. After saving the conf file restart the MYSQL server, `sudo systemctl restart mysql`

```
ubuntu@dbserver:~$ sudo vi /etc/mysql/mysql.conf.d/mysqld.cnf
ubuntu@dbserver:~$
ubuntu@dbserver:~$ sudo systemctl restart mysql
ubuntu@dbserver:~$ sudo systemctl status mysql
● mysql.service - MySQL Community Server
   Loaded: loaded (/usr/lib/systemd/system/mysql.service; enabled; preset: enabled)
   Active: active (running) since Sun 2026-08-30 13:28:34 UTC; 6s ago
     Invocation: eebe6961202b4cb49bf32fdcf54e6efb
   Process: 2584 ExecStartPre=/usr/share/mysql/mysql-systemd-start pre (code=exited, status=0/SUCCESS)
    Main PID: 2595 (mysqld)
      Status: "Server is operational"
        Tasks: 35 (limit: 627)
     Memory: 498.7M (peak: 498.8M)
        CPU: 846ms
      CGroup: /system.slice/mysql.service
              └─2595 /usr/sbin/mysqld

Aug 30 13:28:33 dbserver systemd[1]: Starting mysql.service - MySQL Community Server...
Aug 30 13:28:34 dbserver systemd[1]: Started mysql.service - MySQL Community Server.
ubuntu@dbserver:~$
```

8. Tried connecting from App Server machine, the port is reaching but need to create a user and allow access to remote.

```
ubuntu@appserver:~$ mysql -u db_user -p -h 10.0.20.176 -P 3306
Enter password:
ERROR 1130 (HY000): Host '10.0.10.188' is not allowed to connect to this MySQL server
ubuntu@appserver:~$ telnet 10.0.20.176 3306
Trying 10.0.20.176...
Connected to 10.0.20.176.
Escape character is '^]'.
Host '10.0.10.188' is not allowed to connect to this MySQL server
Connection closed by foreign host.
ubuntu@appserver:~$ telnet 10.0.20.176 330
```

```
mysql> Create user 'sam'@'10.0.10.188' Identified by '123456'
-> ;
Query OK, 0 rows affected (0.01 sec)

mysql> GRANT ALL PRIVILEGES ON *.* TO 'sam'@'10.0.10.188';
Query OK, 0 rows affected (0.01 sec)

mysql> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

9. After granting permission, I have tried to login again in the mysql server via the App Server machine and it was successful.

```
ubuntu@appserver:~$ mysql -u db_user -p -h 10.0.20.176 -P 3306
Enter password:
ERROR 1045 (28000): Access denied for user 'db_user'@'10.0.10.188' (using
password: NO)
ubuntu@appserver:~$ mysql -u sam -p -h 10.0.20.176 -P 3306
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 14
Server version: 8.4.10-0ubuntu0.26.04.1 (Ubuntu)

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statem
ent.

mysql> show databases;
+-----+
| Database |
+-----+
| information_schema |
| mysql |
| performance_schema |
| sys |
+-----+
4 rows in set (0.02 sec)

mysql> use mysql ;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql>
```

♥100% Ethernet * 24°C Hyderabad, Telangana

Therefore, I successfully created a 3-Tier Architecture in the AWS, using the VPC, PUBLIC & PRIVATE SUBNETS, ROUTE TABLES, NAT GATEWAY, ELASTIC IP.